

Experiment No. : 3

Title: Implement control structures and form validation. Create a grading system based on user-entered marks.

2. Software/Tools Required: Visual Studio Code (VS Code) Google Chrome/web browser, HTML, JavaScript.

3. Theory:

A. Form Validation

Ensures user input is accurate and valid before processing.

- Empty Check: Ensures the input field is not blank.
- Type Check: Ensures input is a number using `isNaN()`.
- Range Check: Restricts marks within valid bounds.

B. Control Structures (if...else if...else)

Executes specific blocks of code depending on whether conditions evaluate to true or false. Used to map mark ranges to Grades.

C. DOM Manipulation

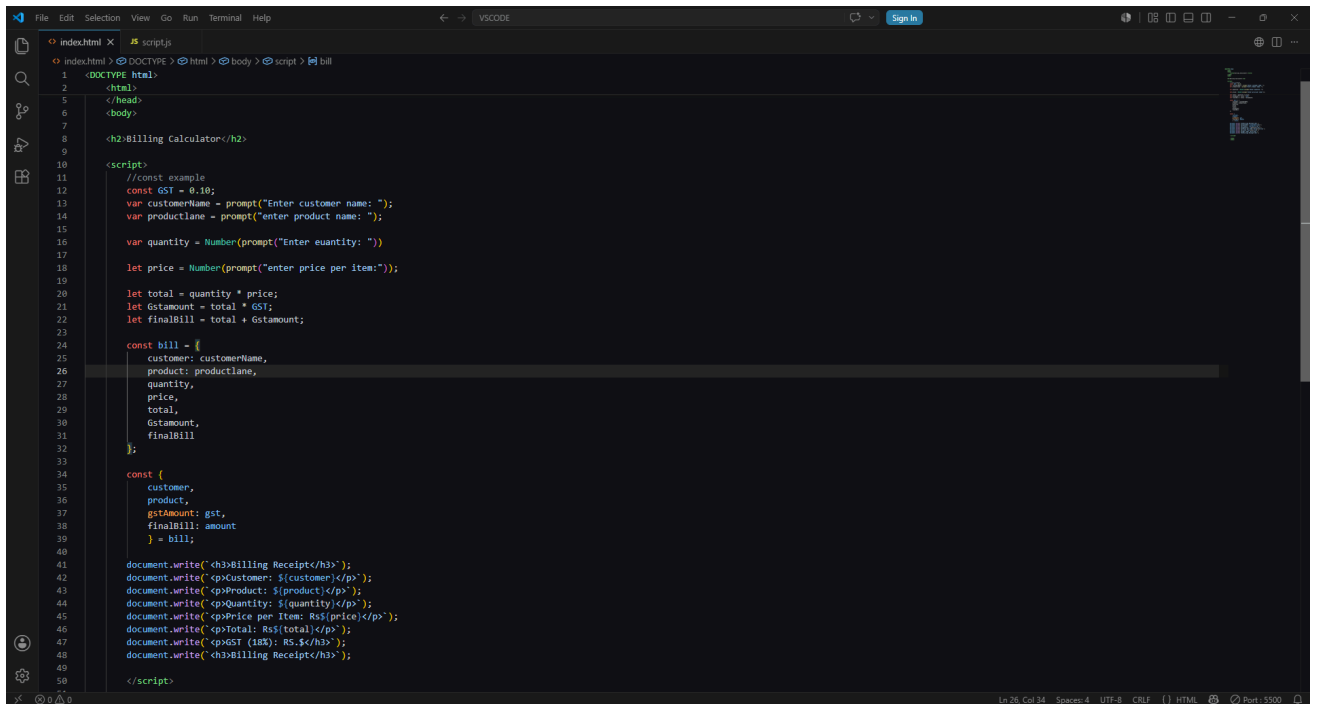
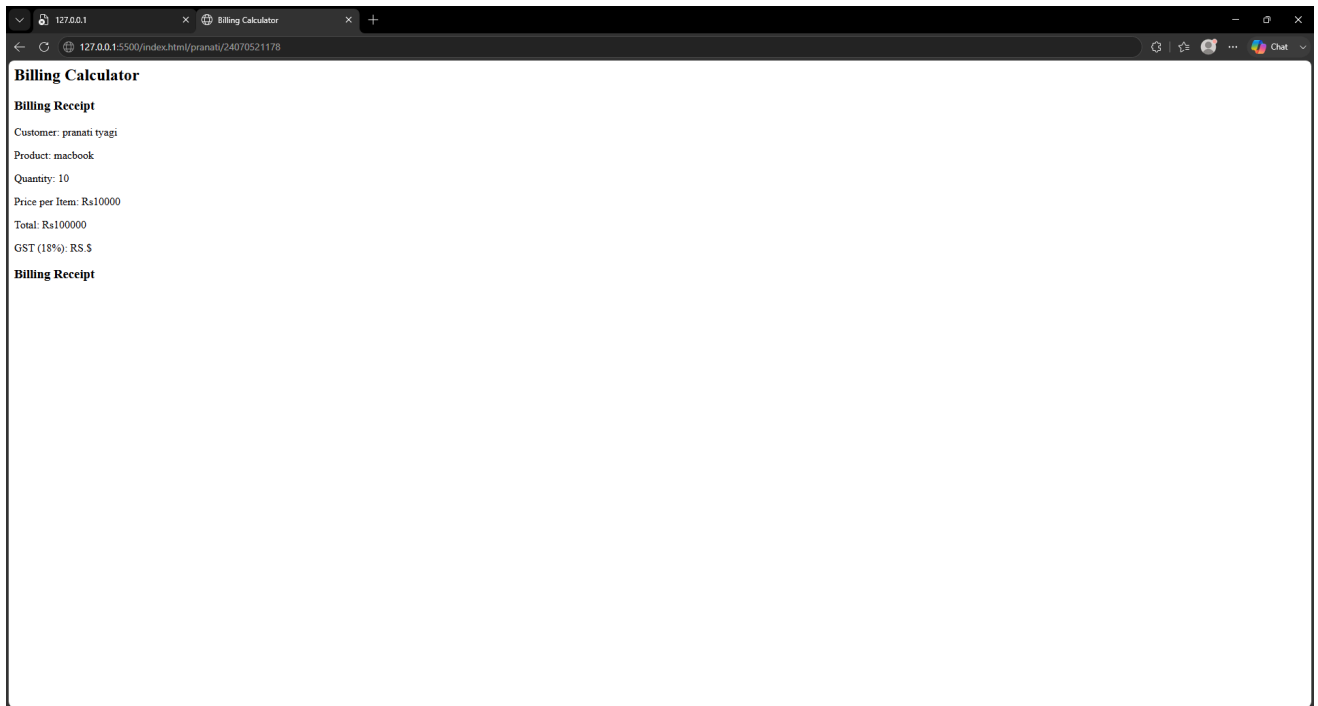
- Input:

`document.getElementById('marks').value` (Reads input).

- Output:

`document.getElementById('result').innerText` (Displays output).

Output:



5. Result/Conclusion: By using control structures (if...else if) alongside form validation techniques, web applications can accurately process user inputs, prevent faulty data processing, and produce dynamic, real-time results such as student grading reports.

